Reg. No.: …… Name: ……..

ALBERTIAN INSTITUTE OF SCIENCE & TECHNOLOGY

Diploma First Internal Examination (September 2025)

Program Code and Semester: AM S3

Course Code and Name: 3341 DISCRETE MATHEMATICS

Max. Marks: 20		**Duration: 1 Hours**		
Q. No	**Question**	**Marks**	**CO**	**BL**
Part A **Answer all questions, each question carries 1 mark**				
1.	If p is a proposition, find $\neg(\neg p)$.	1	CO1	U
2.	If a set A contains n elements, find the number of subsets.	1	CO2	R
Part B **Answer all questions, each question carries 3 marks**				
3.	Write the converse, inverse and contrapositive of the statement: If a number is divisible by 4, then it is even.	3	CO1	A
4.	Given $A = \{1, 2, 3, 4, 5\}$ $B = \{3, 4, 5, 6, 7\}$ Find $A \cup B$, $A \cap B$ and $A - B$.	3	CO2	U
Part C **Answer any 2 questions, each question carries 6 marks**				
5.	Construct the truth table for $p \rightarrow (q \wedge r)$.	6	CO1	A
	OR			
6.	Determine whether $(p \wedge q) \rightarrow p$ is a tautology, contradiction, or contingency.			
7.	Draw Venn diagrams for the following : (i) $A \cap B \cap C$ (ii)$(A \cup B)'$ (iii)$A' \cap B'$	6	CO2	A
	OR			
8.	In a class of 50 students, 28 students like Mathematics and 32 students like Computer Science. If every student likes at least one of the two subjects, find the number of students who like both Mathematics and Computer Science. Draw a suitable two-set Venn diagram and solve using the inclusion–exclusion principle.			